

ZIXUAN (RAY) ZHOU

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EDUCATION

MSE in Financial Mathematics | Johns Hopkins University | Baltimore, MD | 2024-2026

Bachelor of Economics | Central University of Finance and Economics | Beijing, China | GPA: 3.9/4.0, 2.3% | 2020-2024

INTERNSHIP EXPERIENCE

Quantitative Researcher | Broadway Yintai Assets Management Co., Ltd | Beijing, China | Summer, 2024

- Factor Mining: Replicated Barra multi-factor model under supervision in Python; constructed 42 third-level factors and compounded them to form upper-level factors; utilized object-oriented programming, parallel computing, and vectorization for speed; Performed validity tests, rescreening, and back-testing; reached a nice excess return annually with a Sharpe ratio over 1
- Portfolio Optimization: Pulled daily market value of ETF options and futures used for hedging; used them to construct portfolios of every industry; created a ratio to indicate 'degree of balance' within all the industry portfolios and calculated optimal allocation of ETF futures for the possibly smallest ratio of it; Coded a Genetic Algorithm to obtain a better local optimal solution than Lingo's.

Document Reviewer | Guosen Securities Co., Ltd | Beijing China | Summer, 2023

- Document Reviewing: Worked in a bond issuing & undertaking group, examined enterprise application materials for issuing bonds.
- ESG Rating: Collected and organized materials, looked for ESG key entries, and rated a corporation in terms of its ESG performance.

RESEARCH EXPERIENCE

Graduate Thesis | Central Bank Monetary Policy Report Sentiment vs. Index Excess Returns | Beijing, China | Feb.2024 - May.2024

- Independent Research: Completed the paper supervised by Dr. Fuwei Jiang and nominated for the Top Bachelor Theses of CUF.
- Data Processing: Web-scraped monetary policy reports, extracted emotional words, and assigned a score to each report.
- Empirical Analysis: Adopted OLS to estimate the baseline regression, employed different days of excess returns and reported Newey-West adjusted t-values; verified baseline robustness via an ARMA-GARCH model; carried out mechanism analysis to explore 3 feasible channels, employed various proxies of intermediate factors, and included a horse racing between 2 channels.

PROJECT EXPERIENCE

Predicting CAR by CNN over Conference Earning Call | Machine Learning | Baltimore, MD | Apr.2025 - May.2025

- CAR calculation: Obtained historical data. Calculating the CAR (cumulative abnormal return) on different rolling windows. On each window, the log-normal return of individual stocks, as well as the S&P 500, was obtained. Calculated the expected return based on CAPM. Cumulated the abnormal return of various observing windows and labeled CARs for training.
- Transcript processing: Utilized several methods to extract information from the raw conference earnings call transcripts. Filtered out less important words; used a word corpus to extract the emotional words; used Word2Vec to embed the processed texts.

Replication of Option Value | Stochastic Process | Baltimore, MD | Apr.2025 - May.2025

- Monte Carlo Simulation: Created 1000 daily simulations of paths of futures for 365 days, with assumed parameters, entered to buy delta futures at the beginning, each subsequent day adjusted position according to delta, and accumulated profit until expiration. Calculated the replication cost on each simulation path and plotted a histogram for replication cost.
- Different Hedging Strategy: Changed hedging frequency from low to high, observed the mean and standard deviation of hedging cost. Repeated the procedure with different implied volatilities for both more conservative and riskier replication of the option.

3/2 Volatility Model Revisit | Stochastic Process | Baltimore, MD | Apr.2025 - May.2025

- Data Collection: Collected SPX call and put option data, with daily price, volume, open interests, and BS implied volatility.
- Improving codes: Improved analytical codes by spotting and removing unstable intervals, resulting in more reasonable outcomes.
- Others: Performed stress tests of the 3/2 model, calculated Greeks of the model, and plotted errors from crude MC and Mixing MC.

LEADERSHIP

Associate Leader | Media Department of Campus Student Association | Beijing, China | 2021-2023

- Team Management: Led a team of 10 to operate CSA's official media, posting news and reports; arranged college-level activities.
- Volunteering: Led a team of community volunteers; led a team to oversee the lighting of the graduation ceremony stage.

SKILLS CERTIFICATES & HONORS

- Programming: Python / SQL / MATLAB / LaTeX / GitHub
- Certificates: Coursera "Machine Learning Specialization", FRM (Financial Risk Manager) Level I, CFA Level I candidate
- Honors: KPMG 2023 Innovation Paper Prize, Second prize of the Beijing College Student Mathematics Competition, Finance Department Scholarship of CUF, Third prize in the College Students Innovation & Entrepreneurship Competition
- Courses: Machine Learning, Data Structures & Algorithms, Stochastic Process, Probability & Statistics, Linear Algebra, Calculus